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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Byte force
Selected Domain	Domain 2
Selected SDG Track (1–17)	**SDG 8, Target 8.2:** Achieve higher levels of economic productivity through diversification, technological upgrading, and innovation.
Project / Solution Name	InboxPilot
Name (Leader)	HUANG JINYIN
Email & Phone (Leader)	Jinyinhuang18@gmail.com
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	DUAN YINGSHU, DAN ZHEYU, SU MINEN, JIANG CHENJUN

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized enterprises (SMEs) often struggle to manage work-related messages across multiple platforms. Customer service staff and project managers may receive messages from different applications while handling several projects simultaneously, making it difficult to identify which messages require immediate attention.

As a result, important messages can be buried under routine communications, leading to delayed responses and missed deadlines. Repetitive inquiries also consume valuable time, reducing employees' productivity. More critically, urgent issues may go unnoticed because many teams lack an effective system to automatically prioritize messages and escalate unresolved problems.

These challenges can negatively affect customer satisfaction, team efficiency, and business performance. Our team aims to address this problem by developing an **Intelligent Cross-Platform Message Management and Priority Escalation System** that organizes messages, identifies their urgency, and helps ensure that critical issues receive timely attention.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our system integrates messages from multiple software platforms into a single website, allowing users to manage all their communications in one place.

The website categorizes messages by recipient and prioritizes them based on their urgency and importance. Users can manually adjust the importance of individual messages and assign star ratings to senders to increase the priority weight of messages received from those senders.

Users can continuously process messages by clicking the **"Next Item"** button, which presents messages in priority order. The AI automatically generates suggested replies, but human confirmation is required before sending. Only when the risk is assessed as extremely low will the AI automatically send a pre-approved template that has been manually edited and approved by the user.

If an important message remains unprocessed for more than 10 minutes, the system automatically escalates it to the user's supervisor for further action.

The website also maintains records of previously processed messages for tracking and reference. The database is simulated using ChatGPT-generated data. We imported the GitHub repository into Replit and ran the Streamlit workbench through the Run workflow; Preview returned HTTP 200. The app has not been published on Replit. Streamlit Community Cloud remains the intended public host; public access has not been verified.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies, Frameworks, and Cloud Hosting

- **Programming Language:** Python
- **Web Framework:** Streamlit
- **Cloud Hosting Platform:** Streamlit Community Cloud (intended; access unverified); Replit Preview (tested; not published)
- **Data Storage:** Synthetic sample data stored in the format implemented by our team.

System Block Diagram

Synthetic Message Data → Message Processing and Priority Classification → Priority Inbox → AI-Generated Reply Suggestions → User Confirmation and Escalation → Streamlit Web Application

The workbench was imported into Replit and run using the Run workflow. Replit Preview returned HTTP 200, but the app has not been published on Replit. Streamlit Community Cloud remains the intended host; public access has not been verified.

The Replit workspace showed the Free plan and no AI credits used. We have not used any official SDC sponsor-provided GenAI API or cloud platform.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Small and medium-sized enterprises (SMEs) receive and process large volumes of information every day. Due to limited staffing, employees often need to handle various types of issues and take on multiple responsibilities. To address this challenge, we proposed developing an AI-powered website that automatically ranks messages by priority. This system helps employees identify urgent tasks, reducing the time spent on manual prioritization and switching between different software platforms. It also helps prevent important messages from being overlooked and delayed responses.

We then proposed several key features, including a unified inbox, priority ranking, AI-generated reply suggestions, and automatic escalation of messages that remain unprocessed beyond a specified time. Finally, we integrated the team's message-processing logic, priority classification system, and web interface to build a unified prototype.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8, Target 8.2:** Achieve higher levels of economic productivity through diversification, technological upgrading, and innovation. Our project supports this target by using AI-powered message prioritization to reduce time spent on manual message sorting and improve employee productivity.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	30 synthetic messages will be used in the initial test to measure the average time required to manually identify and prioritize urgent messages. The baseline time and the number of urgent messages missed will be recorded before evaluating the prototype
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in average time required to identify and prioritize urgent messages, compared with the manual baseline. Priority classification accuracy (%) and urgent-message detection rate (%).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve at least a 20% reduction in average message prioritization time compared with the manual baseline, and aim for at least 85% priority classification accuracy on the labeled synthetic test dataset by the final prototype presentation.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [√] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [√] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [√] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]